

Raw phi1 voltage calibration

RUN

200038_raw-phi1_fixed-0p4-drift

Status: completed; 300 individual PAX readings

MEASUREMENT

Fixed command 0.4 V; 30 blocks of 10 readings. Command written only on initial change.

VOLTAGE GRID

Single fixed command.

PHYSICAL SETUP

Physical position awaiting confirmation; no assumed port

IN1/IN2 physical mapping awaiting confirmation

WHAT IS RECORDED

Requested RP voltage, ASG register readback, full FPGA OUT1 scope trace, explicitly selected IN1/IN2 scope traces, raw PAX records and acquisition times.

WHAT IS NOT ESTABLISHED

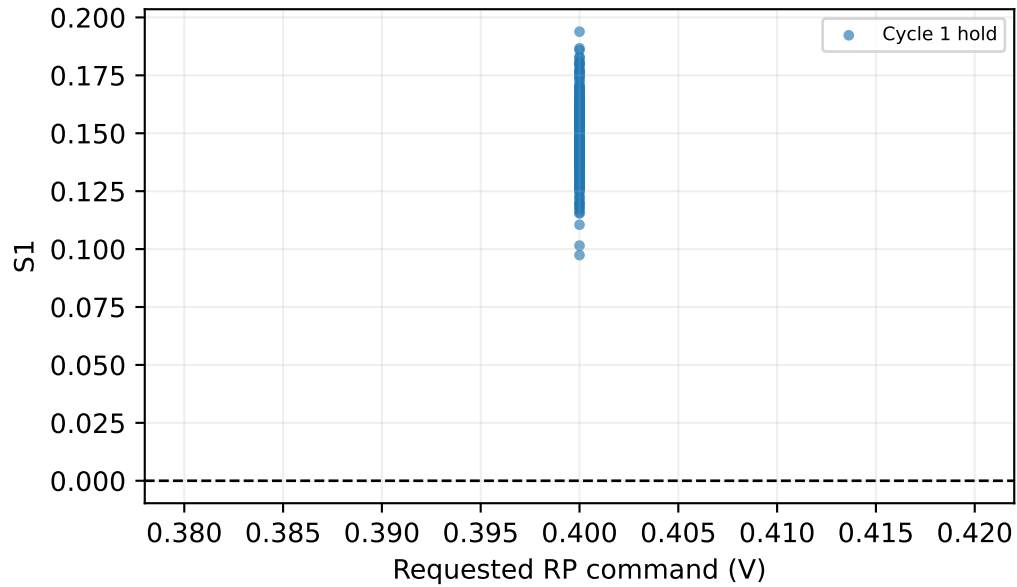
Internal OUT1 data is the signal before the DAC. It is not an independent BNC or piezo-terminal voltage measurement. No historical driver gain is applied.

ANALYSIS POLICY

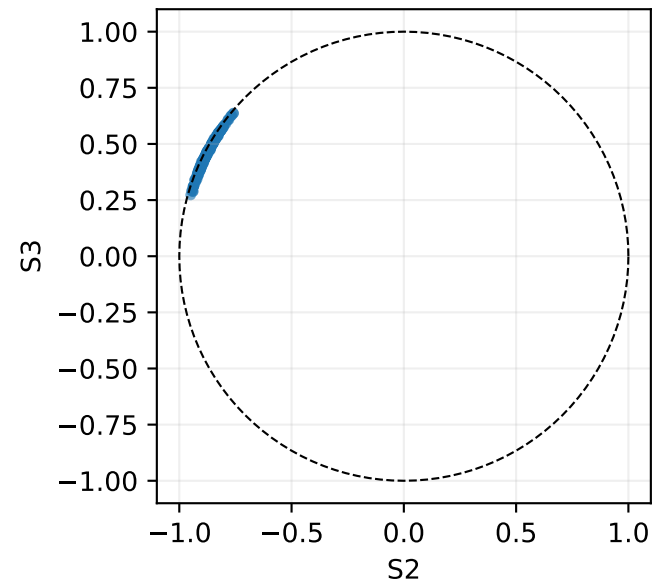
No curve fits, interpolated V_{π} , drift subtraction, or optimized coordinate rotation. Phase is $\text{atan2}(S_3, -S_2)$; unwrapping only adds 2π at coordinate wraps. Step medians are labelled summaries; all individual measurements remain in data.csv and pax_raw.jsonl.

Individual PAX readings at held voltages

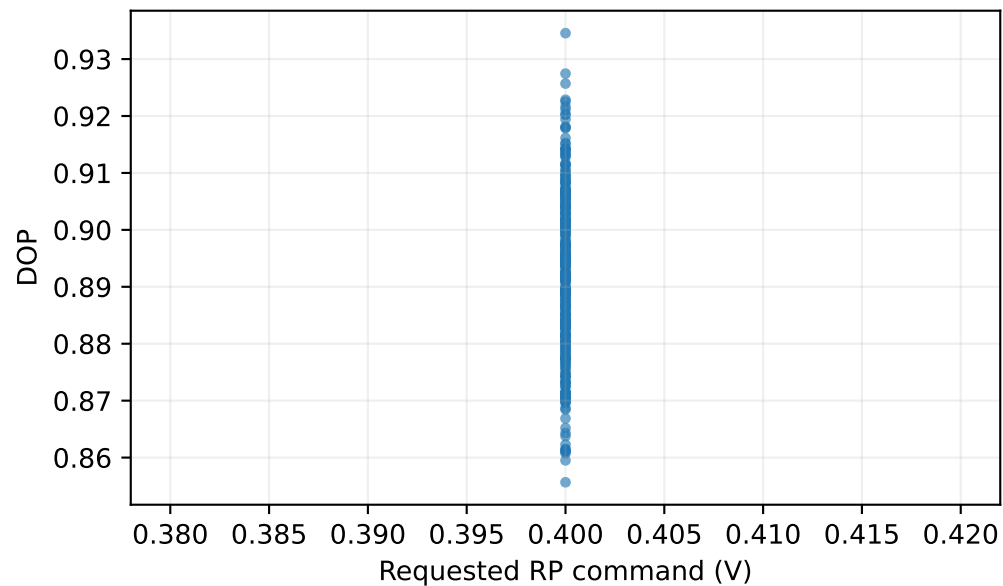
Departure from the $S_1 = 0$ plane



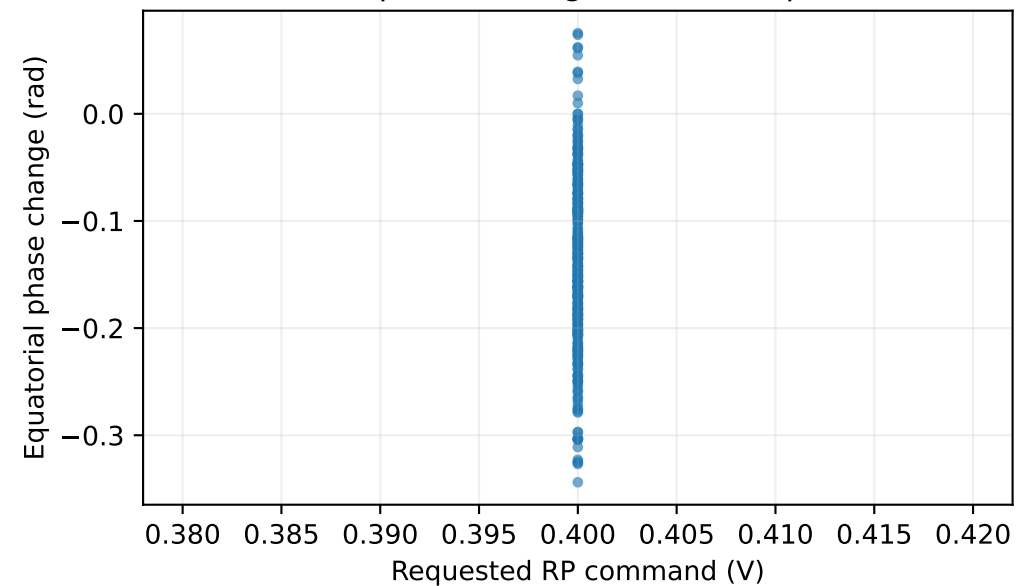
Raw orientation trajectory



Raw PAX DOP



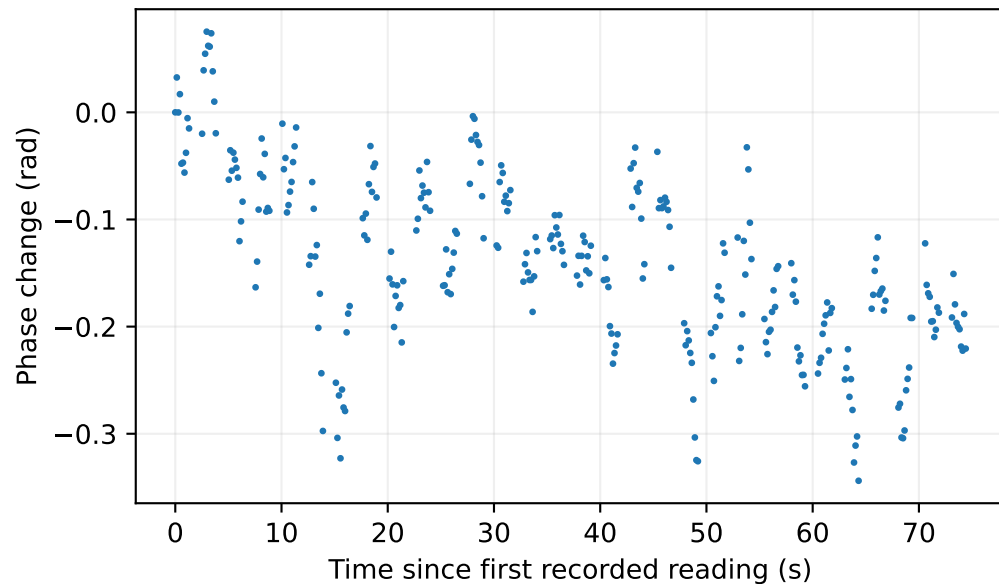
Measured phase change from each pass start



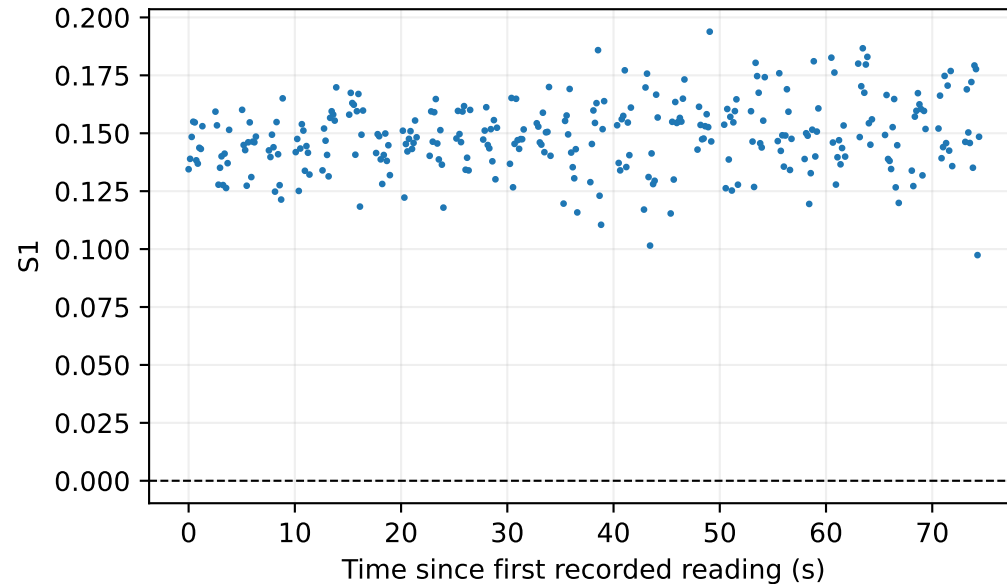
Orientation Stokes are unit-normalized from PAX angles; DOP is shown separately. No fitted or corrected trajectories.

Fixed 0.4 V command: individual measurements over time

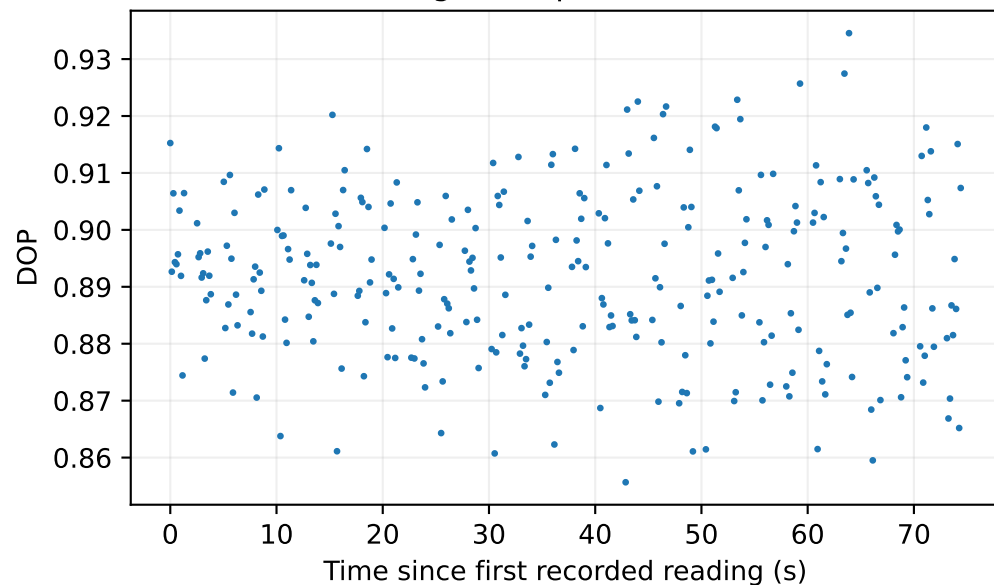
Equatorial phase relative to first reading



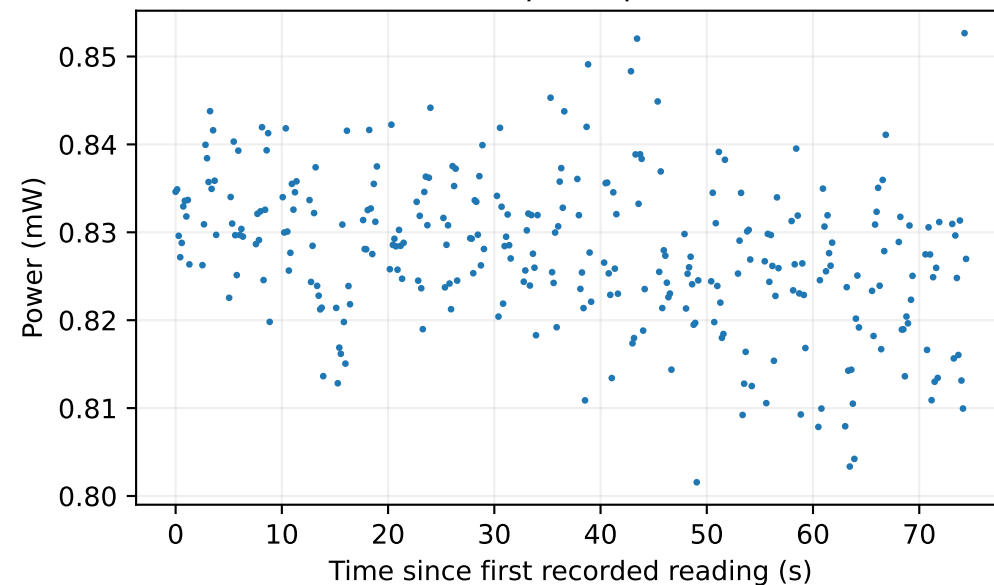
Departure from $S_1 = 0$



Degree of polarization



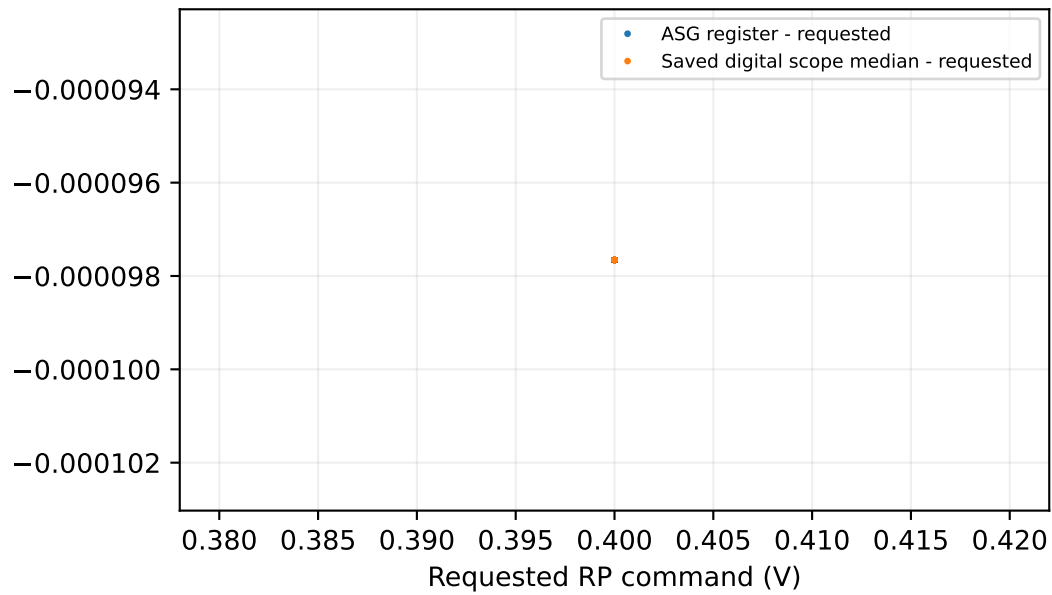
Total optical power



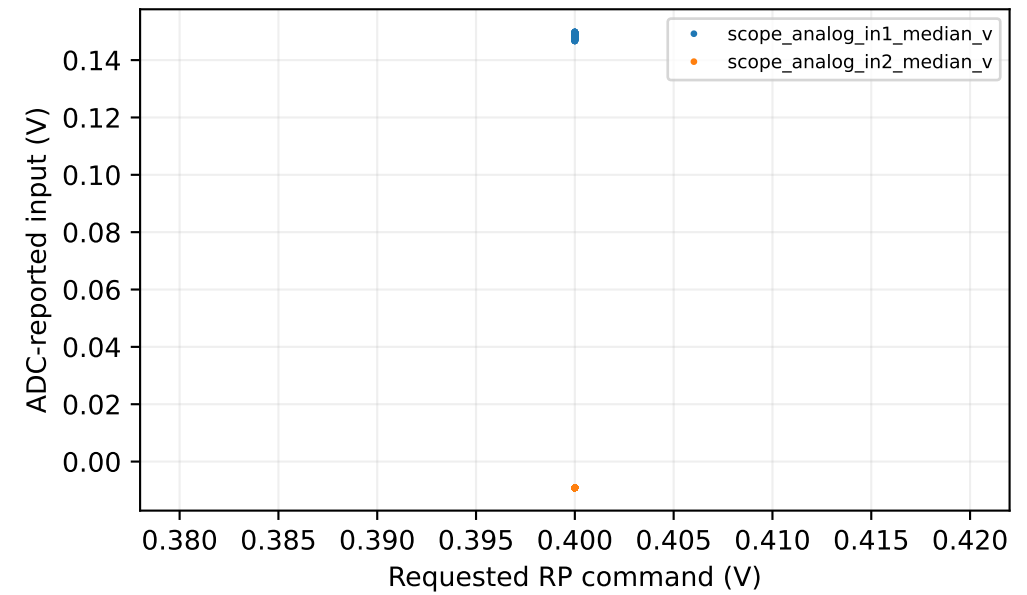
No command changes between blocks. Scatter includes settling, drift and measurement variation; none is subtracted.

Readbacks and direct return-path comparisons

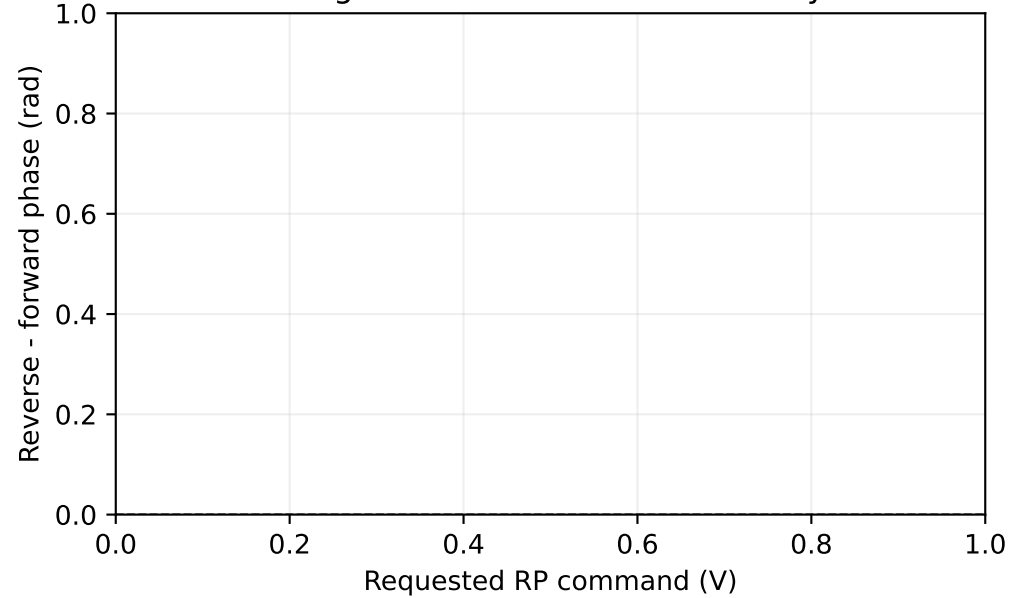
Software command versus internal readbacks



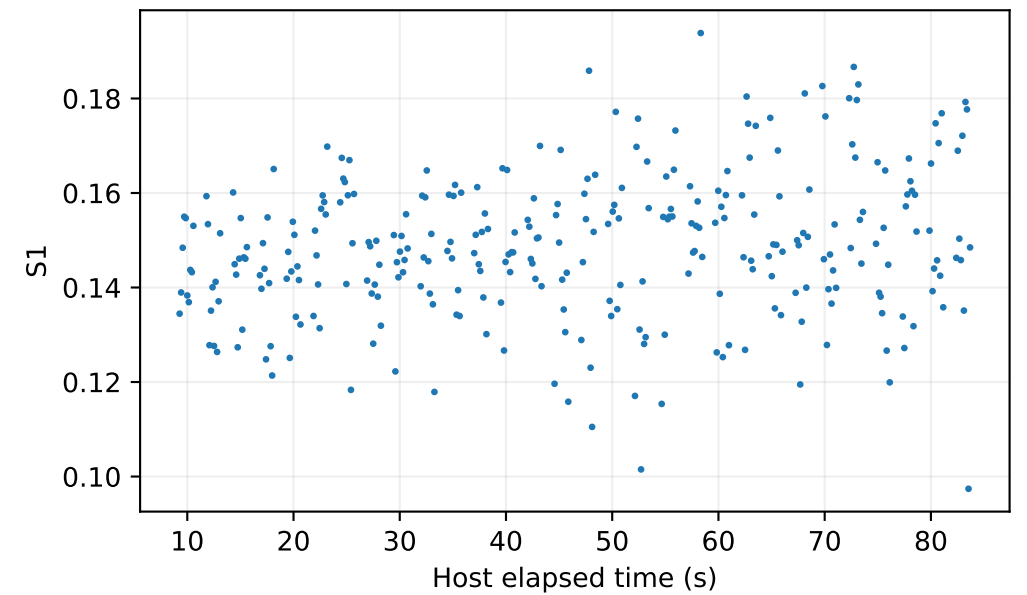
Explicitly routed analog input scope records



Same-voltage return difference: drift + hysteresis



Time record — no drift subtraction



Active acquisition parameters

PROBE

False

START

0.0

STOP

0.8

STEP

0.02

REPEATS

3

LEVELS

None

SETTLE

1.0

SAMPLES

10

HOLD_VOLTAGE

0.4

HOLD_POINTS

30

LABEL

fixed-0p4-drift

Active acquisition parameters

PAX_LOCATION

Physical position awaiting confirmation; no assumed port

INPUT_CONNECTIONS

IN1/IN2 physical mapping awaiting confirmation

PAX SETTINGS

Wavelength 830.0 nm; daemon mode 5; configured rotation 60.0 Hz

CLEANUP EVIDENCE

[0.0, 0.0]